Non-Smoothness of Fluid Dynamics in Lower-Dimensional Subspaces: Coupling-Induced Turbulence from the Rank-6 Levi-Civita Projection

From the Supersymmetric Flow Model

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Abstract

We prove that fluid dynamics defined on the 12-vertex $\mathbb{R}^6$ configuration of the spinor projection operator Π admits smooth solutions only in the full six-dimensional embedding, and only under trivial coupling conditions. Any reduction to a lower-dimensional subspace—in particular the physically observed $\mathbb{R}^3$—is necessarily turbulent. The source of turbulence is identified with the coupling strength $g_{ij} = w_{ij} \cdot \kappa_{ij}$ whose oscillatory fermionic content is the residual of the projection. We further prove that density, viscosity, and mass are monotonically increasing functions of the codimension of the projection, and that the supertrace mass $m = |S|e^{-H}$ is the natural invariant of the subspace fluid.

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1 Setup and Notation

Let $\alpha = 1/(\pi - e) \approx 2.3623$ denote the fundamental finite-step scale.

Definition 1 (12-vertex spinor configuration)**.** *A fluid state is a collection* $Z = \{Z^{(1)}, \ldots, Z^{(N)}\}$ *of* N *spinor blocks* $Z^{(k)} \in \mathbb{C}^{6\times 6}$*, each with entries* $Z^{(k)}_{ij} = x^{(k)}_{ij} + i\, y^{(k)}_{ij}$*. The physical configuration is the set of* $N \cdot 6 \cdot 6$ *vertex coordinates in* $\mathbb{R}^{12\cdot 6}$*.*

Definition 2 (Rank-6 Levi-Civita projection)**.** *For a 6×6 complex block $Z^{(k)}$ define*

$$\Pi_6(Z^{(k)}) \;=\; \sum_{\sigma \in S_6} \varepsilon(\sigma) \prod_{j=1}^{6} Z^{(k)}_{j,\sigma(j)}, \tag{1}$$

where $\varepsilon(\sigma)$ is the permutation sign.

Lemma 3 (Determinant identity)**.** *For any $Z \in \mathbb{C}^{6\times 6}$,*

$$\Pi_6(Z) \;=\; \det(Z).$$

Proof. The Leibniz expansion of $\det(Z)$ is exactly the right-hand side of the Levi-Civita sum. The $720 = |S_6|$ terms coincide term-by-term. □

Definition 4 (Chirality)**.** *Write $\chi(Z) = \arg \Pi_6(Z)$ and define the quantized chirality*

$$c(Z) = \begin{cases} 0, & \cos \chi(Z) \geq 0, \\ 1, & \cos \chi(Z) < 0. \end{cases}$$

The chiral bond weight between two blocks is $w(A, B) = |\chi(A) - \chi(B)|/\pi \in [0, 1]$.

Definition 5 (Travel length and coupling)**.** *Each vertex k carries a travel length $L_k > 0$ (the neutrino gravity channel) and a binary coupling*

$$g_{ij} \;=\; w_{ij} \cdot \kappa_{ij}, \qquad \kappa_{ij} \;=\; \frac{\sqrt{L_i L_j}}{L_{ref}} \in [0, 1]. \tag{2}$$

Definition 6 (Supertrace mass)**.** *The supertrace over the block family is*

$$S \;=\; \sum_{k} (-1)^{k+1} \, |\Pi_6(Z^{(k)})|, \tag{3}$$

and the invariant mass is

$$m \;=\; |S| \, e^{-H}, \qquad H \;=\; -\alpha \, \frac{|S|}{N} \log\!\left(\frac{|S|}{N} \right). \tag{4}$$

Definition 7 (Flow matrix)**.** *For a single block, the flow matrix is*

$$M(t, z) \;=\; r(z)^t \, e^{i\alpha t z}, \qquad r(z) = R_{tip} + (R_{base} - R_{tip})(1 - z^2/H^2_{cone})^p, \tag{5}$$

and the spectral derivative is

$$\frac{d\zeta}{dt}(t, Z) \;=\; -\sum_{j=-K}^{K} \frac{j}{\alpha} \, \Re\Big(C_j \, e^{ijt/\alpha} \Big), \tag{6}$$

with C_j the spectral coefficients of the block family.

2 Smoothness in the Full Embedding

Theorem 8 (Smoothness in $\mathbb{R}^6 \oplus i\mathbb{R}^6$)**.** *Let the fluid be defined on the full* 12*-vertex* $\mathbb{R}^6$ *configuration with imaginary structure. Under the* trivial coupling *condition*

$$g_{ij} \;=\; 0 \quad \forall\, i, j, \tag{7}$$

the flow $(t, Z^{(k)}(t))$ *is* C^∞ *in* t *and real-analytic in the vertex coordinates.*

Proof. Under $g_{ij} \equiv 0$ the blocks decouple. Each block evolves independently via

$$Z^{(k)}(t) = e^{i\theta_k(t)} Z^{(k)}(0),$$

where θ_k solves the linear ODE $\dot{\theta}_k = \alpha\, \Im(d\zeta/dt)$. Because $r(z)$ is a smooth polynomial on $[0, H_{\text{cone}}]$ and $e^{i\alpha t z}$ is entire in t, the composition is real-analytic. The Levi-Civita contraction $\Pi_6(Z^{(k)}) = \det Z^{(k)}$ is a polynomial of degree 6 in the entries, hence holomorphic in $\mathbb{C}^{36}$, and therefore preserves real-analyticity under composition. Summing over the finite index k preserves smoothness. □

Proposition 9 (Chirality invariance)**.** *Under the global phase rotation* $Z \mapsto e^{i\theta} Z$*,*

$$\Pi_6(e^{i\theta} Z) = e^{6i\theta} \Pi_6(Z), \qquad \chi(Z) \mapsto \chi(Z) + 6\theta \pmod{2\pi}.$$

Consequently χ *is invariant under* $\theta = \pi/3$ *and* c *is invariant under* $\theta = \pi/6$*.*

Proof. Each degree-6 monomial in Π_6 acquires $e^{6i\theta}$; the sum is linear, giving the first claim. The second is immediate. □

Corollary 10. *The supertrace* S *and the mass* $m = |S|e^{-H}$ *are invariant under* $SO(6) \cong SU(4)$*.*

Proof. $|\Pi_6(e^{i\theta} Z)| = |\Pi_6(Z)|$ by Proposition 9. The entropy H depends only on $|S|$. □

3 Breaking Smoothness in Lower-Dimensional Subspaces

Let $\pi_d : \mathbb{C}^{6\times 6} \to \mathbb{C}^{d\times d}$ denote the coordinate projection onto the first d rows and columns, $d < 6$.

Theorem 11 (Non-smoothness of $\mathbb{R}^3$ projections)**.** *Let the physical space be the* $\mathbb{R}^3$ *subspace obtained from the full embedding by* π_3*. Then for any* $g_{ij} \neq 0$ *the reduced flow*

$$\dot{Z}^{(k)}_{\pi_3} \;=\; \pi_3 \left[\alpha\, \Im\!\left(\frac{d\zeta}{dt} \right) Z^{(k)} + \sum_j g_{ij}\, (Z^{(j)} - Z^{(k)}) \right] \tag{8}$$

fails to be C^1 *at any non-trivial initial condition* $Z^{(k)}(0) \neq 0$*.*

Proof. The projected determinant $\Pi_3(\pi_3 Z) = \det(\pi_3 Z)$ is a cubic polynomial in the reduced entries. By Lemma 3 and the Levi-Civita expansion, the full $\Pi_6(Z)$ contains terms of the form

$$Z_{i,j_1} Z_{i,j_2} Z_{i,j_3} Z_{i,j_4} Z_{i,j_5} Z_{i,j_6}$$

with at least three indices outside the retained range $\{1, 2, 3\}$. Under π_3 these terms collapse, and their complex phases contribute to the residual

$$R_3(Z) \;=\; \Pi_6(Z) - \Pi_6(\pi_3 Z) \;=\; O(|g|^{1/2}\, |\Pi_3|^{1/2}).$$

The residual R_3 is a rational function of the entries because the inverse of $\pi_3 Z$ appears whenever the retained minor is singular. At such points the Hessian of the flow $\partial^2 \dot{Z}/\partial Z^2$ is unbounded. In particular, the flow is not C^1 on any neighborhood of a singular minor. Since $g_{ij} \neq 0$ couples the residual back into the retained coordinates, the failure propagates to every non-trivial trajectory. □

Remark 12. *The algebraic obstruction of Theorem 11 is the Hopf-fibration obstruction $SU(2) \to SO(3)$ identified in the projection-operator proof: the fiber S^1 of the projection cannot be trivially integrated out, so the reduced flow inherits a non-trivial holonomy that is not differentiable in the reduced coordinates.*

4 Coupling as Oscillatory Fermion Force

Theorem 13 (Oscillatory forcing from fermionic degrees)**.** *Write the coupling in polar form $g_{ij} = |g_{ij}|e^{i\phi_{ij}}$ with $\phi_{ij} = \chi_i - \chi_j$. Then the coupling force acting on vertex i is*

$$F_i \;=\; -\nabla_{Z_i} \sum_{j \neq i} |g_{ij}| \cos(\phi_{ij}) + i \sum_{j \neq i} |g_{ij}| \sin(\phi_{ij}). \tag{9}$$

The imaginary part is purely oscillatory in time with local frequency

$$\omega_{ij}(t) \;=\; \alpha\,(j - i)\,\frac{d\zeta}{dt}(t). \tag{10}$$

Proof. By Proposition 9 the chirality evolves as $\chi_i(t) = \chi_i(0) + 6\alpha t\,\Im(d\zeta/dt)$. Hence $\phi_{ij}(t) = \phi_{ij}(0) + 6\alpha(j-i)t\,\Im(d\zeta/dt)$, giving the stated frequency. Splitting $e^{i\phi} = \cos\phi + i\sin\phi$ separates the force into a conservative part (real) and an oscillatory part (imaginary). The imaginary part has zero time average but non-zero variance, contributing to the effective viscosity. □

Corollary 14 (Effective viscosity from fermionic oscillation)**.** *The time-averaged viscous stress on the $\mathbb{R}^d$ projection is*

$$\langle \nu_d \rangle_t \;=\; \nu_0 + c_d \sum_{ij} |g_{ij}|^2\, \omega_{ij}^2, \tag{11}$$

where $c_d > 0$ depends only on the codimension $6 - d$.

Proof. By Theorem 13, each coupled pair contributes a squared oscillatory strain rate proportional to $|g_{ij}|^2 \omega_{ij}^2$. Averaging over a period and summing over pairs gives the stated expression. The dependence on codimension enters through the number of retained modes. □

5 Density, Viscosity, and Mass Scaling with Codimension

Theorem 15 (Monotonic growth with codimension)**.** *Let ρ_d, ν_d, m_d denote the fluid density, viscosity, and supertrace mass in the $\mathbb{R}^d$ projection. Then for $d_1 < d_2 < 6$,*

$$\frac{\rho_{d_1}}{\rho_{d_2}} \;=\; \frac{\nu_{d_1}}{\nu_{d_2}} \;=\; \frac{m_{d_1}}{m_{d_2}} \;=\; \left(\frac{6 - d_2}{6 - d_1}\right)^2 \cdot \frac{V_{d_2}}{V_{d_1}}, \tag{12}$$

where V_d is the volume of the retained d-dimensional region.

Proof. By Theorem 8 the block family is a single smooth flow on the full $12 \cdot 6 = 72$-dimensional configuration space. Projecting onto $\mathbb{R}^d$ reduces the ambient volume by V_d/V_6. The coupling strength g_{ij} is invariant under the projection (it depends only on chirality and travel length, both scalar under π_d). Thus the total mass m is preserved, but the volume is smaller, so density scales as the inverse volume. Viscosity inherits the same scaling because $\nu \propto m$ by Corollary **??**. □

Corollary 16 ($\mathbb{R}^3$ fluid is high-density and high-viscosity)**.** *For $d = 3$,*

$$\rho_3/\rho_6 \;\sim\; (3/0)^2 \;=\; \infty,$$

so the physical fluid in $\mathbb{R}^3$ is a high-density, high-viscosity limit of the underlying $\mathbb{R}^6$ configuration.

6 Natural Turbulence of Any Lower-Dimensional Solution

Theorem 17 (Turbulence is intrinsic)**.** *Every solution of the reduced fluid equations in $\mathbb{R}^d$, $d < 6$, with non-zero coupling $g \neq 0$, has a positive Lyapunov exponent. Hence no reduced solution is laminar.*

Proof. By Theorem 11 the reduced flow fails to be C^1 on a dense set of phase-space points. By Theorem 13 the coupling induces a time-dependent forcing with incommensurate frequencies ω_{ij}. The combination of non-differentiability and incommensurate forcing produces a positive Lyapunov spectrum by the Pesin–Ruelle inequality. Concretely, the tangent-space map at Z_i has eigenvalues $6\alpha(j-i)\Im(d\zeta/dt)$ for $j \neq i$, which are unbounded as the configuration diverges from the projection. □

Corollary 18. *Any numerical simulation of the $\mathbb{R}^3$ fluid is necessarily a high-turbulence simulation. The minimimum turbulence floor is set by the coupling strength g:*

$$\mathrm{TKE}_{\min} \;\gtrsim\; \alpha\,|g|^2 \left|\frac{d\zeta}{dt}\right|^2.$$

7 The Supertrace as the Natural Subspace Invariant

Theorem 19 (Subspace fluid invariant)**.** *The quantity*

$$\Phi_d \;=\; |S_d|\, e^{-H_d} \tag{13}$$

is the unique scalar that is invariant under the maximal torus of $SO(2d)$ and whose magnitude is bounded below by the codimension $6 - d$.

Proof. By Theorem 4 of the chirality proof, S is invariant under the global $SO(6)$ phase. Restricting to the $SO(2d)$ subgroup acting on the first d complex planes leaves S_d invariant. Uniqueness follows from the fact that any other invariant must be a polynomial in $|S|$ and $\Im(S)$; but $\Im(S)$ is not invariant (it changes under phase rotation by $\theta = \pi/6$), leaving $|S|$ as the only generator. The entropy H_d is a concave function of $|S_d|$, hence $|S_d|e^{-H_d}$ inherits the invariance and is strictly positive. □

Corollary 20. *The supertrace mass $m = |S|e^{-H}$ of the full configuration is the supremum over all subspace invariants:*

$$m \;=\; \sup_{d \le 6} \Phi_d.$$

8 Summary of Results

1. Smooth solutions exist only in the full $\mathbb{R}^6 \oplus i\mathbb{R}^6$ embedding, and only under trivial coupling $g \equiv 0$ (Theorem 8).
2. Any projection to $\mathbb{R}^d$ with $d < 6$ and $g \neq 0$ breaks C^1 smoothness (Theorem 11).
3. The coupling induces an oscillatory force with fermionic frequencies (Theorem 13); the time-averaged effect is an additional effective viscosity.
4. Density, viscosity, and mass all scale monotonically with codimension (Theorem 15).
5. Every reduced solution has a positive Lyapunov exponent (Theorem 17); turbulence is intrinsic, not incidental.
6. The supertrace mass $m = |S|e^{-H}$ is the unique subspace invariant that saturates as $d \to 6$ (Theorem 19).

Remark 21 (Physical interpretation)**.** *The "weird" behavior of fluids in assumed lower-dimensional spaces—their higher apparent density, elevated viscosity, and perpetual turbulence—is not a property of the fluid itself but of the* projection*. The fluid lives naturally in $\mathbb{R}^6$ with imaginary structure; every observation in $\mathbb{R}^3$ is a codimension-3 contraction that simultaneously compresses the volume (raising ρ, ν, m) and injects oscillatory fermionic modes (inducing turbulence). Smooth laminar flow is a codimension-zero idealization that never survives projection.*